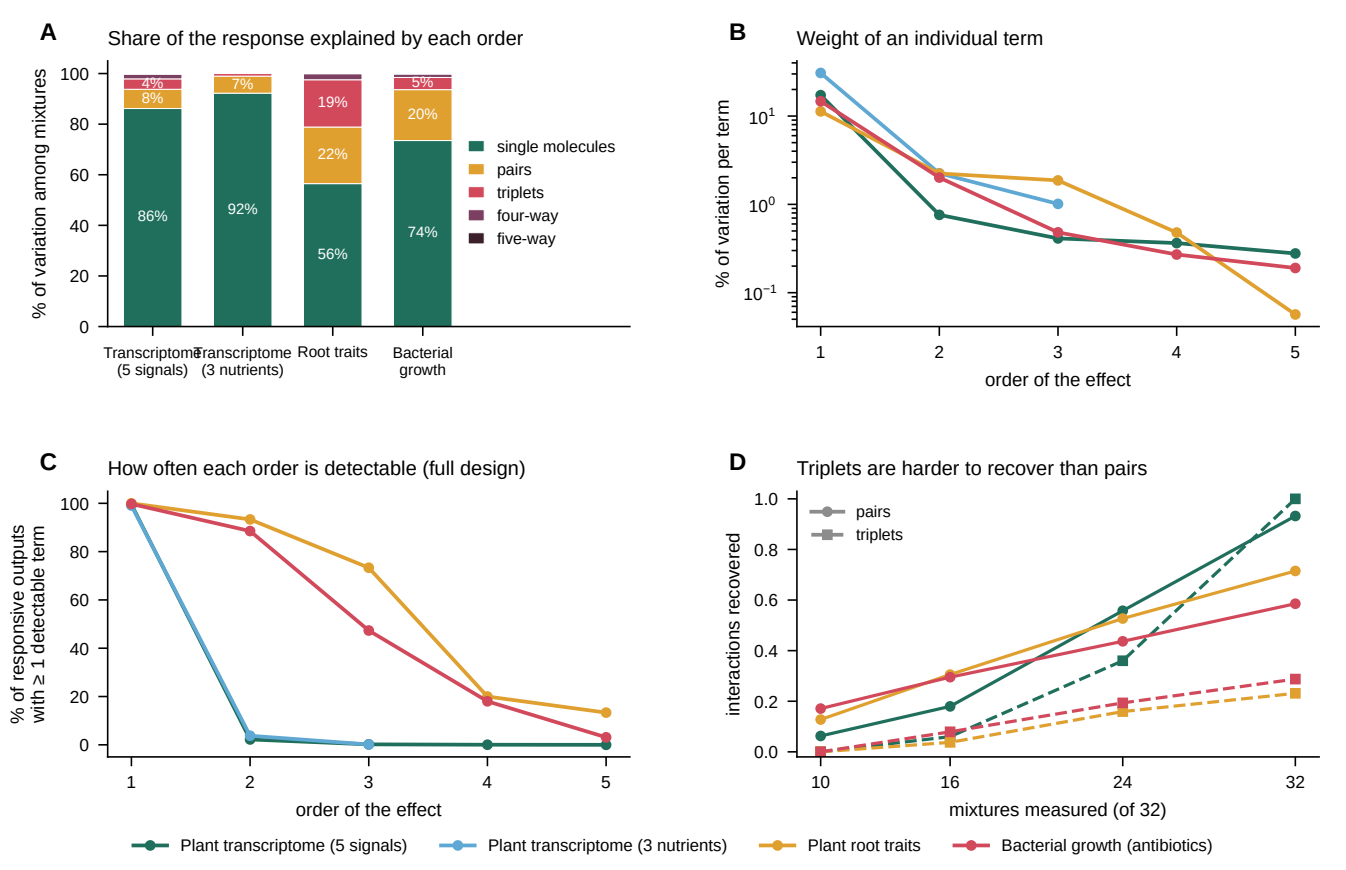
A
Share of the response explained by each order
% of variation among mixtures
100
80
60
40
20
0
4%
8%
86%
7%
92%
19%
22%
56%
5%
20%
74%
single molecules
pairs
triplets
four-way
five-way
Transcriptome (5 signals)
Transcriptome (3 nutrients)
Root traits
Bacterial growth
B
Weight of an individual term
% of variation per term
$10^1$
$10^0$
$10^{-1}$
1
2
3
4
5
order of the effect
C
How often each order is detectable (full design)
% of responsive outputs with ≥ 1 detectable term
100
80
60
40
20
0
1
2
3
4
5
order of the effect
D
Triplets are harder to recover than pairs
interactions recovered
1.0
0.8
0.6
0.4
0.2
0.0
pairs
triplets
10
16
24
32
mixtures measured (of 32)
Plant transcriptome (5 signals)
Plant transcriptome (3 nutrients)
Plant root traits
Bacterial growth (antibiotics)